

Semiconductor diodes

Multiple Choice Questions for one mark

1. The arrow direction in the diode symbol indicates.....

- a) Direction of electron flow.
- b) Direction of hole flow (Direction of conventional current)**
- c) Opposite to the direction of hole flow
- d) None of the above

2. The knee voltage (cut in voltage) of Si diode is.....

- a) 0.2 V
- b) 0.7 V**
- c) 0.8 V
- d) 1.0 V

3. When the diode is forward biased, it is equivalent to.....

- a) An off-switch
- b) An On-switch**
- c) A high resistance
- d) None of the above

4. Under normal reverse bias voltage applied to diode, the reverse current in Si diode.....

- a) 100 mA
- b) order of μA**
- c) 1000 μA
- d) None of these

5. Avalanche breakdown in a diode occurs when.....

- a) Potential barrier is reduced to zero.
- b) Forward current exceeds certain value.
- c) Reverse bias exceeds a certain value.**
- d) None of these

6. Reverse saturation current in a Silicon PN junction diode nearly doubles for very.....

- a. 20 rise in temp.
- b. 50 rise in temp.
- c. 60 rise in temp.
- d. 10 rise in temp.**

7. A forward potential of 10V is applied to a Si diode. A resistance of 1 K Ω is also in series with the diode. The current is.....

- a. 10 mA
- b. 9.3 mA**
- c. 0.7 mA
- d. 0 Ma

$$I_d = (10 - 0.7) / 1000$$

8. When a reverse bias is applied to a diode, it will.....

- a. Raise the potential barrier**
- b. Lower the potential barrier
- c. Increases the majority-carrier a current greatly
- d. None of these

9. The best description of zener diode is that

- a. **it operates in reverse region** b. it is a constant voltage device
- c. it is a constant current device d. none of the above

10. The LED is usually made of.....

- a. GeSi b. C and Si **c. GaAs** d. none of the above

11. Testing a good diode with an ohmmeter should indicate

- a. high resistance when forward or reverse biased
- b. low resistance when forward or reverse biased
- c. high resistance when reverse biased and low resistance when forward biased**
- d. high resistance when forward biased and low resistance when reverse biased

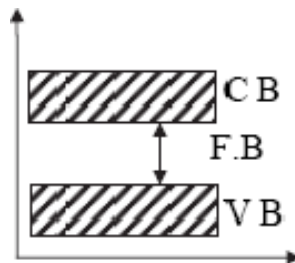
12. A P-N junction allows current flow when.....

- a. both the n-type and p-type materials have the same potential
- b. the n-type material is more positive than the p-type material
- c. the p-type material is more positive than the n-type material**
- d. there is no potential on the n-type or p-type materials

13. The diode used in seven segment display is.....

- a. zener diode b. Photo diode **c. LED** d. LASER diode

14. The diagram shown below corresponds to,



- (a) the single energy level of an electron.
- (b) the discrete energy level of an electron.
- (c) the energy transfer diagram.

(d) the energy band diagram.

15. The forward biased resistance of the diode is than its reverse biased resistance.

- (a) larger (b) double **(c) smaller** (d) none of these

16. A zener diode is operated in

- (a) breakdown region** (b) forward characteristics region
- (c) zero biasing (d) none of these

17. If the p-n junction diode is heavily doped then breakdown voltage will

- (a) increases **(b) decreases** (c) remains same (d) none of these

18. The barrier potential for an unbiased silicon junction diode at room temperature is.....

- a. 0.2 V **b. 0.7 V** c. 0.8 V d. 1.0 V

19. A zener diode is used as a.....

- (a) half wave regulator (b) half wave rectifier
- (c) voltage regulator** (d) amplifier

20. The acronym LED stands for

- (a) light energized diode **(b) light emitting diode**
- (c) low energy device (d) low energy dynamo

21. When the diode is forward biased, it is equivalent to

- a. An off-switch **b. An On-switch** c. A high resistance d. None of the above

22. By adding.....impurity in intrinsic semiconductor P type semiconductor is made.

- a. trivalent** b. pentavalent c. quadra valant d. divalent

23. The charge on P-type semiconductor is.....

- a. positive **b. neutral** c. negative d. either positive or negative

n type is also neutral

Bipolar Junction Transistors

- 1) A transistor hasPN junctions.
a. one **b. two** c. three d. four
- 2) The emitter isdoped.
a. heavily b. lightly **c. moderately** d. not
- 3) The base isdoped.
a. heavily **b. lightly** c. moderately d. not
- 4) The collector is.....doped.
a. heavily b. lightly c. moderately d. not
- 5) The value of α is _____.
a. less than 1 b. greater than 1 c. less than 0 d. equal to 0
- 6) The main function of a transistor is to do.....
a. rectification **b. amplification** c. light emission d. heat emission
- 7) Transistors would be classified as.electronic devices.
a. active b. passive c. both active and passive d. neither active nor passive
- 8) From working of transistor operation one can write _____.
a. $I_B = I_C + I_E$ b. $I_C = I_B + I_E$ c. $I_E = I_C - I_B$ **d. $I_E = I_C + I_B$**
- 9) For CE transistor configuration o/p characteristics is graph of _____.
a. I_B verses V_{BE} b. I_B verses V_{CE} c. I_E verses V_{CE} **d. I_C verses V_C**